Sorting: Optional for only the interested

- Advanced optimizations that pipeline operators:
- SELECT uid, experience FROM Users ORDER BY experience

  - At write in Pass 0 to disk (if needed):
    - Do projection on necessary attributes → each tuple is smaller → less pages
    - When you write to disk you can fill pages at 100% → even less pages
    - that is, projection and compacting pages is combined with the sort!
    - Example:
      - User has 40.000 tuples 500 data
      - Write of pass 0:
        - » 8 Bytes (uid, experience) * 40.000 / 4000 Bytes = 80 pages
      - Pass I has to read in much less

Projection Distinct Operation

```
SELECT   DISTINCT uname
FROM     USER
```

- Basic project often done on the fly (as discussed previously)
- More complex: SELECT DISTINCT name FROM …
  - Requires sort in order to eliminate duplicates!!
  - Often done at the very end (less tuples) or whenever the relation is sorted for some other reason
  - Expensive operation: use DISTINCT only when really necessary

Set Operations

- Intersection and cross-product special cases of join.
- Union (Distinct) and Except similar;
- For instance: sorting based approach to union:
  - Sort both relations (on combination of all attributes).
  - Scan sorted relations and merge them.

Aggregate Operations (AVG, MIN, etc.)

- Usually done at the very last step after all selections/joins etc.
- Without grouping:
  - In general, requires scanning the relation.
- With grouping:
  - Sort on group-by attributes, then scan relation and compute aggregate for each group. (Can improve upon this by combining sorting and aggregate computation.)